Deterministic Edge ML Architecture for Real-Time Friction Estimation

(μ_used proxy based derived from multi-sensor vehicle data)

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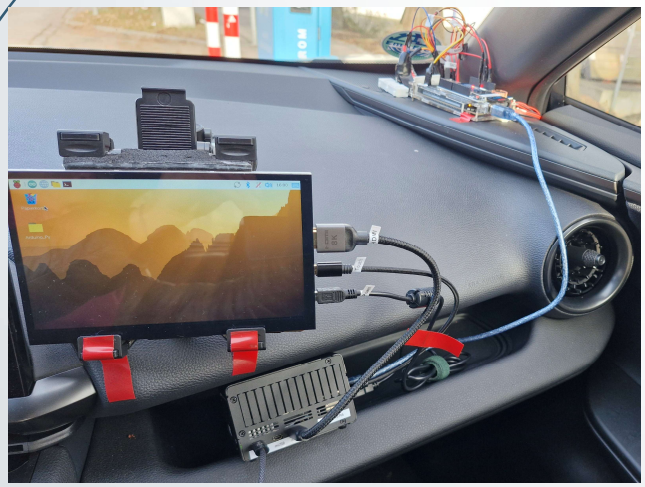

Sensors: Multi-Modal Sensor Input Layer
- IMU (a_long, ay, az, gyro) @ 100 Hz
- OBD2 (v, RPM, Throttle) @ 100 Hz
- (optional) Camera (Vision) @ 30 Hz
- Ambient environmental sensors (T, Humidity, p) @ 1 Hz
- Timestamp synchronisation (UTC based)

Sample-based Buffer: Deterministic Sample-based Data Buffer
- Fixed control Interval : 100ms (10 Hz)
- Resampling asynchroner sensors
- Ring Buffer architecture
- Plausibility checks and outlier rejection
- Worst-Case Latency control

Feature Extraction: Real-Time Feature Engineering Layer
- Jerk computation (da/dt)
- RMS calculation over sliding window
- Normalization (Min-Max/ Z-Score)
- Feature vector dimension: n=4

ML Model Layer: Deterministic Linear Model
- Model: Ridge Regression
- Feature dimension: n = 4
- Training: LORO-CV (Leave-One-Run-Out)
- Regularization parameter: a = 1.0
- Offline validation RMSE (avg): 1.23e-4, Offline validation R^2 (avg) 0.999998
- Model size: ~ 10 kB ; Inference complexity: O(n)

Edge Inference: CPU-based deployment
- Platform: Raspbery Pi5 (ARM Cortex-A76)
- Execution mode: CPU based linear inference
- Fixed control cycle: 100 ms (10 Hz)
- Deterministic sample based loop
- Real-time μ_used estimation

μ_pred: Real-Time μ_used Estimation
- Continuous μ_used prediction
- Update Rate: 10 Hz
- Deterministic output per control cycle
- Based on longitudinal vehicle dynamics

Logging: Embedded Performance & Resource Monitoring
- Latency logging, CPU utilization, RAM usage
- Temperature monitoring, Power estimation

Deterministic Online Inference Loop

Cycle definition

- Fixed control cycle: 10 ms (10 Hz)
- Deterministic multi-sensor sampling
- Worst case latency bounded (< 10 ms)
- Closed form linear inference

Online Processing Steps (pro Zyklus)

1. Sensor read (IMU, OBD2, BME, RacheChronoPro)
2. Resampling & buffer update
3. Feature computation (jerk, RMS, normalization)
4. Linear ML inference (Ridge regression, CPU)
5. μ_pred output update
6. Performance logging

Time budget

Schritt	Value*
Feature extraction	~0.05 ms
Linear inference (CPU)	~0.6 ms (mean)
Full cycle (worst case)	~2.1 ms (max)
Total latency	~2.5 ms (< 10 ms)

**estimated for Raspberry Pi 5 (ARM Cortex-A76) using conservative 5x scaling from x86 validation measurements*

Determinism Strategy

- Fixing 100 Hz scheduling
- Pre-allocated memory
- No dynamic model loading
- Ring-buffer based sendor handling
- Closed-form Ridge regression ensures bounded execution time

Hardware Awarness (current implementation)

Compute Platform

- X86 validation baseline (development platform)
- Raspbery Pi 5 (ARM Cortex-A76)
- Linux-based embedded environment

Model Design & Optimization (current implementation)

- Linear Ridge regression baseline
- Closed-forn inference (O(n))
- Regularization parameter tuning (a=1.0)
- LORO-CV (Cross-validation Leave-One-Run-Out)
- Validation under deterministic cycle constraints

Ressource Metrics

Metric	Value*
Mean interference time	~0.6 ms
Worst case latency	~2.1 ms
Model size	1.33 kB
RAM footprint	~130MB

**estimated for Raspberry Pi 5 (ARM Cortex-A76) using conservative 5x scaling from x86 validation measurements*

Design Trade-offs

- Acuuracy vs latency
- Model simplicity vs interpretability
- Determinism vs adaptive scheduling

Energy consideration

- Thermal monitoring under sustained load
- CPU Load impact analysis
- Scalability toward automotive SoC

Key Architectural Contributions

Deterministic Edge AI Architecture

- 100 Hz fixed control loop with bounded worst-case latency
- closed-form linear inference (O(n)) without dynamic graph execution
- pre-allocated memory and ring-buffer based data handling
- deterministic multi-sensor synchronization

Hardware Aware Deployment Strategy

- cross- architecture validation (X86 → ARM Cortex A76)
- quantified inference latency (mean & worstcase)
- explicit timing budget verification (<10 ms constraint)
- ressource footprint analysis (model size, RAM usage)

Scalability Perspective

- Architecture transferable to automotive-grade SoCs (CPU/NPU based architecture)
- Accelerator-ready design (e.g. Coral TPU)
- Separation offline optimization and deterministic runtime execution